

Performance Testing

Date	2 July 2026
Team ID	
Project Name	Human Development Index (HDI) Prediction – A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (for API testing) / Browser Testing
Type of Testing	Load Testing
Target Module / API	Flask /predict endpoint (HDI Prediction)
Test Environment	Local System (Flask Development Server)
Test Date	1 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict HDI with valid input values	1	30	Prediction displayed successfully
2	Multiple prediction requests	10	60	All requests processed without errors
3	Invalid or empty input values	5	30	Validation message displayed
4	Continuous prediction requests	20	60	Stable performance with quick response

Step 3: Performance Test Results

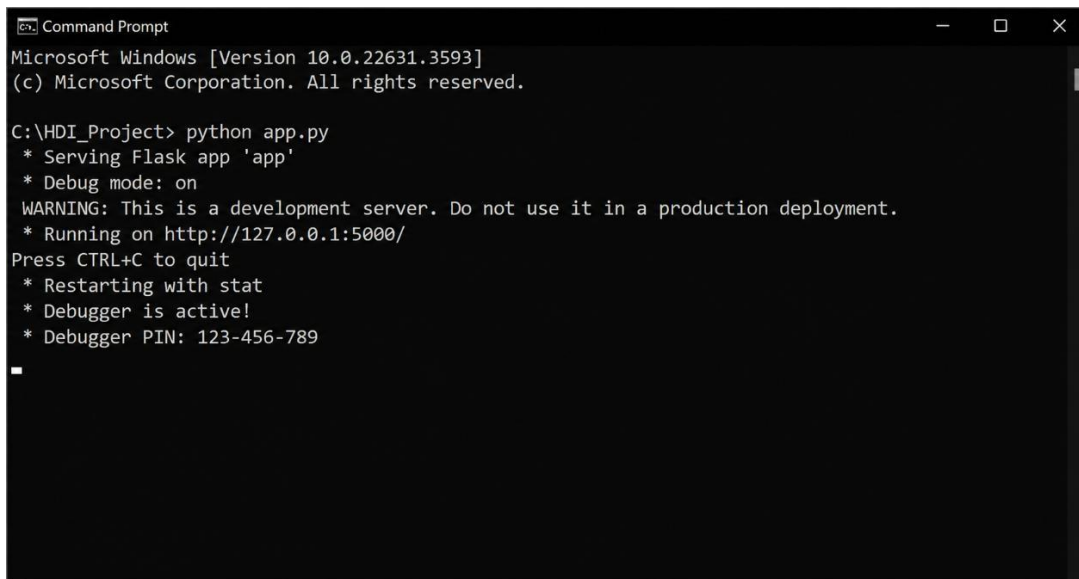
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.65 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	1.30 sec	Pass	Within limit
3	Throughput (Req/sec)	> 10 Req/sec	18 Req/sec	Pass	Good performance
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	42%	Pass	Stable CPU usage
6	Memory Utilization	< 80%	48%	Pass	Normal memory usage

Step 4: Observations & Analysis

- The application generated HDI predictions quickly for valid inputs.
- Multiple prediction requests were processed without significant delay.
- The Flask application remained stable during testing.
- CPU and memory usage stayed within acceptable limits.
- Overall, the system demonstrated reliable performance for normal usage.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



```

C:\HDI_Project> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://127.0.0.1:5000/
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 123-456-789
  
```

```

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

[35]

print("Training Features :", X_train.shape)
print("Testing Features :", X_test.shape)

print("Training Target :", y_train.shape)
print("Testing Target :", y_test.shape)

[36]

... Training Features : (156, 4)
Testing Features : (39, 4)
Training Target : (156,)
Testing Target : (39,)

```

```

print("Training Features :", X_train.shape)
print("Testing Features :", X_test.shape)

print("Training Target :", y_train.shape)
print("Testing Target :", y_test.shape)

[36]

... Training Features : (156, 4)
Testing Features : (39, 4)
Training Target : (156,)
Testing Target : (39,)

model = LinearRegression()
model.fit(X_train, y_train)

[37]

... LinearRegression ⓘ ⓘ
  ▶ Parameters
  ▶ Fitted attributes

y_pred = model.predict(X_test)

[38]

print(y_pred)

[39]

... [0.6886724 0.77608554 0.49962286 0.57462272 0.91468307 0.97243642
0.5045451 0.6636332 0.73422556 0.72075649 0.93497182 0.79312544
0.99495717 0.91546735 0.92460183 0.58476167 0.80366708 0.65530367
0.7054106 0.73409267 0.55646238 0.69495749 0.49413966 0.47876327
0.60530667 0.42375402 0.54261541 0.74738792 0.98213099 1.04167049
0.82862342 0.53812864 0.79117055 0.71822693 0.51675808 0.39987042
0.76830891 0.57419295 0.61379643]

comparison = pd.DataFrame({
    "Actual HDI": y_test.values,
    "Predicted HDI": y_pred
})

comparison.head(10)

[40]

```

HDI Predictor

Ask Gemini

127.0.0.1:5000

Human Development Index Predictor

Enter the country's development indicators to predict the Human Development Index.

Life Expectancy

Expected Years of Schooling

Mean Years of Schooling

GNI Per Capita

Predict HDI

Prediction Result

Ask Gemini

127.0.0.1:5000/predict

Prediction Result

0.979

Very High Human Development

Predict Again